PHYSICS 2111 · PHYSICS NINJA PRACTICE EXAM

Exam 1 · Version A · Solutions

21 questions · 90 minutes · original practice exam, not an official exam

ANSWER KEY

1	**e**	2	**c**	3	**a**	4	**b**	5	**b**	6	**b**	7	**c**
8	**a**	9	**a**	10	**b**	11	**c**	12	**a**	13	**c**	14	**a**
15	**d**	16	**b**	17	**b**	18	**b**	19	**b**	20	**a**	21	**a**

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A car starts from rest and accelerates uniformly at 2.5 m/s² for 6 s. It then moves at constant speed for 10 s, and finally brakes uniformly to a stop in 5 s.

1) What is the maximum speed of the car?

Answer: (e) 15 m/s

$v = at = (2.5)(6) = 15$ m/s.

2) What total distance does the car travel?

Answer: (c) 233 m

Area under v(t): $\frac{1}{2}(15)(6) + (15)(10) + \frac{1}{2}(15)(5) = 45 + 150 + 37.5 = 233$ m.

3) While the car is braking, its acceleration vector points

Answer: (a) opposite to the direction of motion.

The car is slowing down, so a points opposite to v.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A soccer ball is kicked from level ground with an initial speed of 18 m/s at 40° above the horizontal. Ignore air resistance.

4) How long is the ball in the air?

Answer: (b) 2.36 s

$T = 2v_0\sin\theta / g = 2(11.6)/9.81 = 2.36$ s.

5) How far from the kicker does the ball land?

Answer: (b) 32.5 m

$R = v_0\cos\theta \cdot T = (13.8)(2.36) = 32.5$ m.

6) What is the speed of the ball at the top of its path?

Answer: (b) 13.8 m/s

At the top only v_y is zero, so $v = v_x = v_0\cos\theta = 13.8$ m/s.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A stone is thrown horizontally at 8 m/s from the top of a 25 m tall cliff.

7) How far from the base of the cliff does the stone land?

Answer: (c) 18.1 m

$t = \sqrt{(2h/g)} = 2.26$ s, so $x = (8)(2.26) = 18.1$ m.

8) If the stone were thrown horizontally at 16 m/s instead, the time it takes to reach the ground would

Answer: (a) stay the same.

The vertical motion does not depend on the horizontal speed.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A river 80 m wide flows east at 1.5 m/s. A boat moves at 4 m/s relative to the water.

9) The boat points due north. How far downstream does it land?

Answer: (a) 30 m

$t = W / v_{boat} = 20$ s, drift $= v_{water} \cdot t = 30$ m.

10) At what angle west of north must the boat point to land directly across?

Answer: (b) 22 °

$\sin\theta = v_{water} / v_{boat} = 1.5/4$, $\theta = 22°$.

11) Compared with pointing due north, the crossing time when the boat lands directly across is

Answer: (c) longer.

Only the component $v_{boat}\cos\theta$ carries the boat across, so the crossing takes longer.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 4 kg block sits on a frictionless incline at 30°. It is tied by a light string over a frictionless pulley at the top of the incline to a 3 kg block hanging vertically. The blocks are released from rest.

12) What is the magnitude of the acceleration of the blocks?

Answer: (a) 1.4 m/s²

$a = (m_2 g - m_1 g \sin 30°)/(m_1 + m_2) = (29.4 - 19.6)/7 = 1.4$ m/s².

13) What is the tension in the string?

Answer: (c) 25.2 N

$T = m_2(g - a) = 3(9.81 - 1.4) = 25.2$ N.

14) Compared with the weight of the hanging block, the tension is

Answer: (a) smaller.

The hanging block accelerates downward, so the net force on it points down: $T < m_2 g$.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 12 kg box is dragged across a level floor at constant velocity by a rope pulling at 25° above the horizontal. The coefficients of friction are $\mu_s = 0.5$ and $\mu_k = 0.35$.

15) What is the tension in the rope?

Answer: (d) 39.1 N

$T\cos25° = \mu_k(mg - T\sin25°)$, so $T = \mu_k mg/(\cos25° + \mu_k \sin25°) = 39.1$ N.

16) The normal force on the box is

Answer: (b) less than mg.

The rope pulls partly upward, so the floor pushes up less: $N = mg - T\sin25°$.

THE NEXT 2 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A 400 kg roller-coaster car travels upside down at the top of a vertical loop of radius 12 m with a speed of 14 m/s. The car rides on the inside of the track.

17) What is the magnitude of the normal force from the track on the car at the top?

Answer: (b) 2610 N

$N + mg = mv^2/r$, so $N = 400(14^2/12 - 9.81) = 2610$ N.

18) What is the minimum speed at the top for the car to stay in contact with the track?

Answer: (b) 10.8 m/s

Contact is lost when $N = 0$: $v = \sqrt{(gr)} = 10.8$ m/s.

THE NEXT 3 QUESTIONS PERTAIN TO THE SITUATION DESCRIBED BELOW.

A satellite moves in a circular orbit 2000 km above Earth's surface. ($R_E = 6.37 \times 10^6$ m, $M_E = 5.97 \times 10^{24}$ kg, $G = 6.67 \times 10^{-11}$ N·m²/kg²)

19) What is the orbital speed of the satellite?

Answer: (b) 6.9 km/s

$r = 8.37 \times 10^6$ m; $v = \sqrt{(GM/r)} = 6.9$ km/s.

20) What is the period of the orbit?

Answer: (a) 2.12 h

$T = 2\pi r/v = 7620$ s $= 2.12$ h.

21) If the satellite had twice the mass, its orbital speed at the same radius would be

Answer: (a) the same.

The satellite mass cancels: $v = \sqrt{(GM/r)}$.
